

Deepfake Audio Detection:

An Empirical Baseline across Feature Types

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Introduction & Background

- **Task:** Binary classification of bonafide vs. spoofed
- **Dataset:** ASVspoof 2019 LA[1]
- **Main metric:** Equal Error Rate (EER) — lower is better
- **Also used:** Precision, Recall, F1-score

Research Questions:

- Can a shallow CNN achieve **acceptable performance** for the task?
- Which acoustic features **best** detect **spoofing**?
- How do **feature processing** techniques **impact** model **performance**?

Baseline model

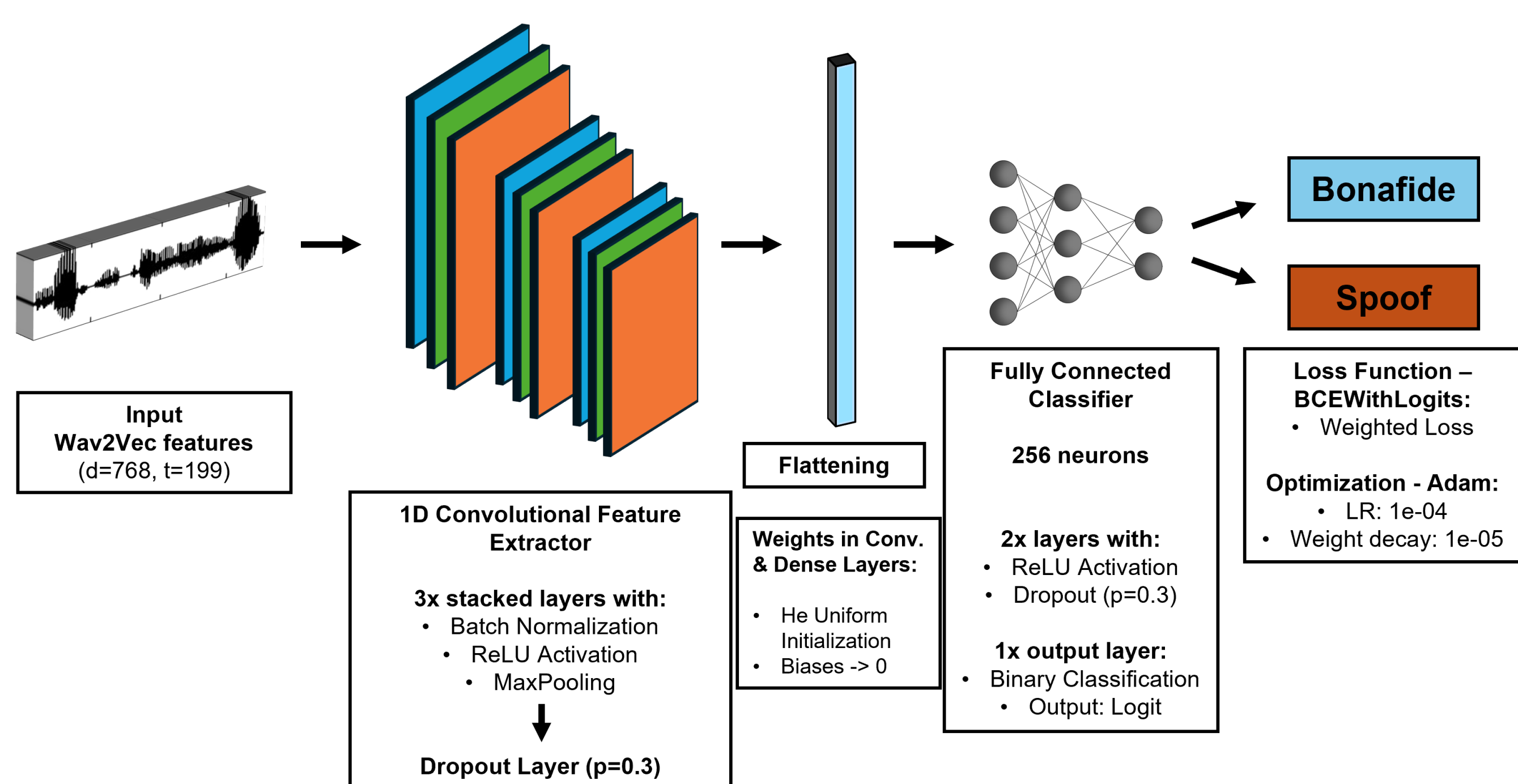


Figure 1: Deeper CNN Model (Wav2Vec features)

Results

The results presented below are based on the full ASVspoof2019 LA dataset: approximately **25k** samples for **training**, **24k** for **development**, and **71k** for **evaluation**.

Feature Type	Model	EER (%)	F1 Score	Accuracy	Prec.	Rec.
Wav2vec	DeepCNN(908K)	3.86	0.83	96.36%	0.74	0.96
Wav2vec	CNN(482K)	4.38	0.82	95.82%	0.72	0.96
LFCC + Δ	CNN(273K)	7.43	0.72	71.43%	0.59	0.93
LFCC	CNN(264k)	8.17	0.69	57.82%	0.56	0.92
LFCC + Δ	DeepCNN(551K)	11.16	0.62	53.29%	0.48	0.89
LogMEL + Δ	CNN(297K)	12.96	0.58	83.59%	0.43	0.87
MFCC + Δ	CNN(273K)	14.48	0.54	90.88%	0.40	0.85
LogMEL	CNN(273K)	16.61	0.51	83.71%	0.37	0.83
MFCC	CNN(264K)	16.81	0.50	90.23%	0.36	0.83
STFT	CNN(359K)	29.27	0.33	71.55%	0.21	0.70

Table 1: Results summary for model architectures and features.

Findings

- **Shallow CNNs with enhancements** → **good performance** in Deepfake audio detection, reaching **EERs < 10%**, while remaining **lightweight and easy to train**.
- **Wav2Vec features** offer the **best detection performance**, but are **high-dimensional** and **require more complex training**.
- Among acoustic features, **LFCC Δ + Δ Δ** achieved the **best results** using Shallow CNN, offering a **good balance of performance and efficiency**.
- **Model-feature compatibility matters:**

Deeper CNN + Rich Features (Wav2Vec) → **Best performance**

Shallow CNN → **Best performance** with Acoustic Features

Deeper CNN + Simple Features → **Worse** than Shallow CNN

Frontend UI & XAI

Initial Objective: Provide a frontend UI in which users can upload an audio file and receive a prediction indicating whether it is bonafide or spoofed.
Extended Goal: Gain deeper insights into the model's behavior and decision-making using **explainable AI (XAI)** techniques.

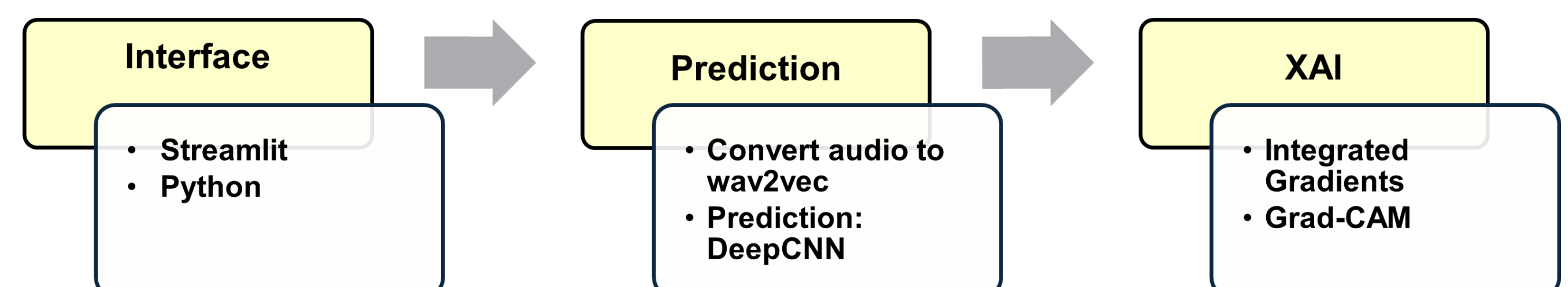


Figure 2: Frontend UI diagram.

Results

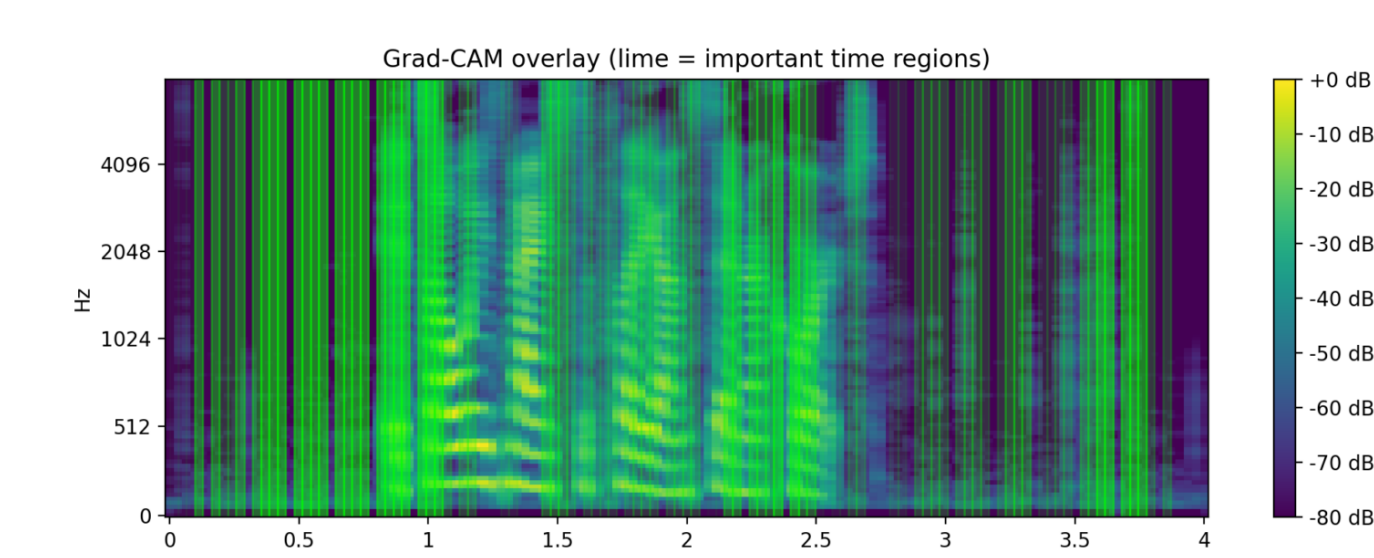


Figure 3: Bonafide audio spectrogram with Grad-CAM overlay.

Findings

- XAI results revealed that the model often relies on **silence**[2] for its predictions—contrary to our expectations.
- Despite good performance on the **test set**, the model misclassified our own bonafide recordings as **spoof**.
- A recording with only **background noise** (no speech) was classified as **bonafide**, aligning with XAI explanations.
- The app loads faster with the **Shallow CNN + LFCC Δ + Δ Δ** model than with the **Deeper CNN + Wav2Vec** model.

Mixture of Experts & LCNN

Initial Objective: The LCNN model is commonly used as a baseline in the challenge. We aim to compare its performance with our self-built model. Additionally, we incorporate an external ("in-the-wild") dataset to train two models and apply MoE to assess performance.

Results

Feature Type	Model	EER (%)	F1 Score	Accuracy	Precision	Recall
LFCC + Δ	LCNN	9.15	0.67	88.82%	0.53	0.91
LFCC + Δ	LCNN & MoE	7.59	0.72	87.49%	0.58	0.92

Table 2: Results of LCNN and LCNN & MoE

Findings

- LCNN achieved an EER of 9% on the evaluation dataset using LFCC features, slightly worse than our baseline. After applying MoE to two trained LCNN models, the EER dropped to 7%.
- Unlike our baseline model, which performed better using BCELogitsLoss for unbalanced data, MoE with LCNN achieved superior results using BCE-CrossEntropy.

Take Home: Shallow, efficient models with features like LFCC can detect deepfake audio well—but explainability shows they may rely on unexpected cues like silence, highlighting the need to pair accuracy with interpretability for real-world reliability. [3][4]

References

- [1] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, H. Delgado, A. Nautsch, J. Yamagishi, N. Evans, T. Kinnunen, and K. A. Lee, "Asvspoof 2019: Future horizons in spoofed and fake audio detection," in *Proc. Interspeech*, 2019, pp. 1008–1012. [Online]. Available: https://www.isca-archive.org/interspeech.2019/todisco19_interspeech.pdf
- [2] Z. Huang, Z. Tang, D. Wang, Z. Wu, and H. Meng, "The impact of silence on speech anti-spoofing," in *Proceedings of Interspeech 2021*, 2021, pp. 1099–1103.
- [3] N. Subramani and D. Rao, "Learning efficient representations for fake speech detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.
- [4] Z. Huang, Z. Tang, D. Wang, Z. Wu, and H. Meng, "The impact of silence on speech anti-spoofing," *arXiv preprint arXiv:2309.11827*, 2023, arXiv:2309.11827.